

Lecture 13

Outliers, Normal Quantiles, Transformations and the Margin of Error

Review

1 What is a density curve?

A smooth curve that describes a distribution; the total area under it is 1.

2 What is the standard normal distribution?

The normal distribution with mean 0 and standard deviation 1, written $Z \sim N(0, 1)$.

3 What is a z-score?

The number of standard deviations a value falls from the mean: $z = (x - \bar{x}) / s$.

4 What is the empirical rule?

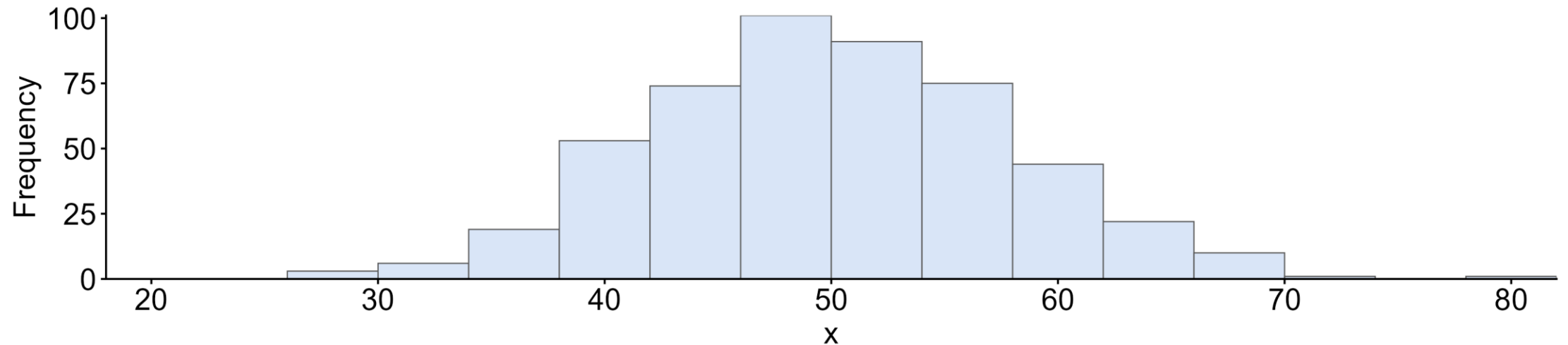
About 68%, 95% and 99.7% of values fall within 1, 2 and 3 SDs of the mean (bell-shaped only).

Review: from x to z

A variable x has the distribution shown below.

$\bar{x} = 50$ $s = 8$

How many standard deviations from its mean is an observation of $x = 62$?



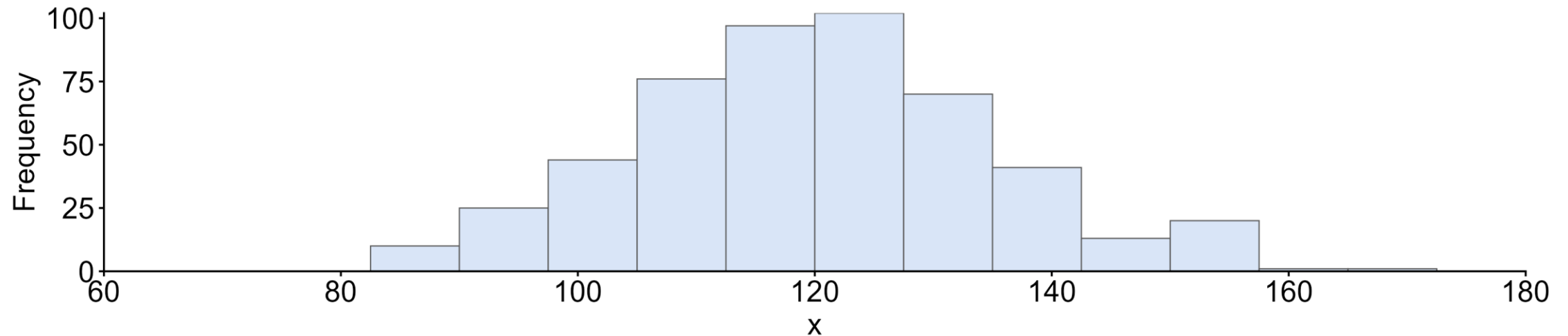
$z = (62 - 50) / 8 = 1.5$: 1.5 standard deviations above the mean

Review: from z to x

A variable x has the distribution shown below.

$\bar{x} = 120$ $s = 15$

What value of x marks 3 standard deviations above the mean?



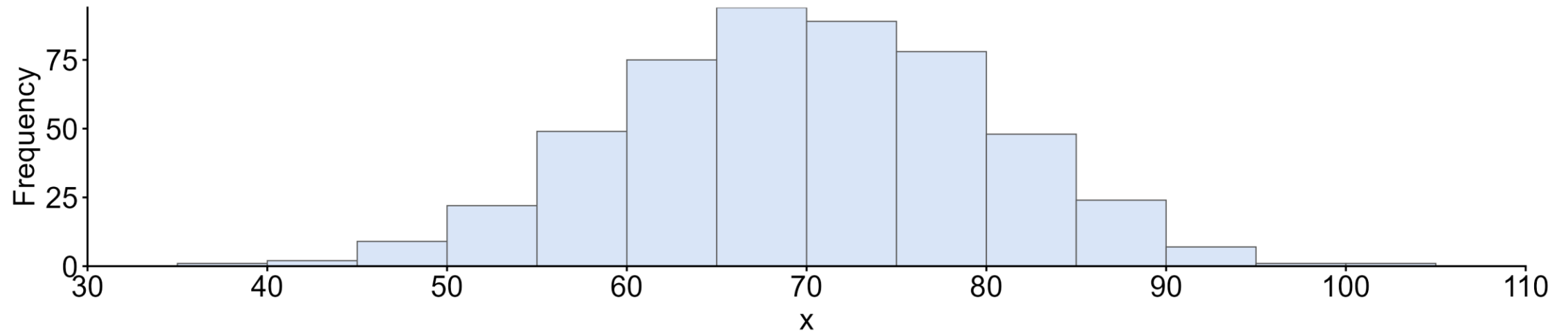
$$x = \bar{x} + z s = 120 + 3(15) = 165$$

Review: the empirical rule

A variable x has the distribution shown below.

$\bar{x} = 70$ $s = 10$

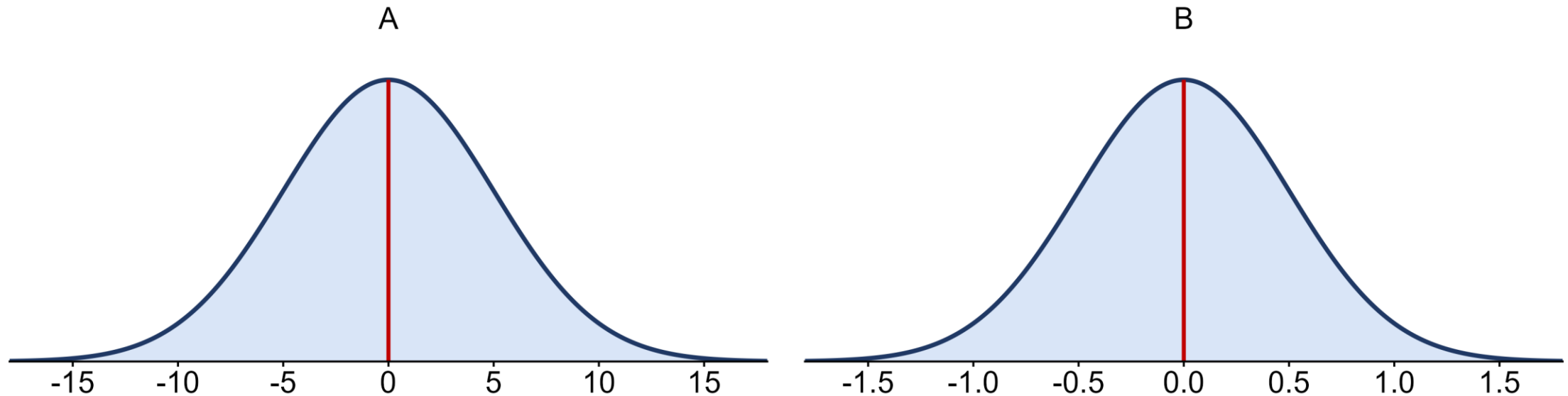
Approximately what value of x is the 16th percentile?



$50\% - 34\% = 16\%$ of values lie below $\bar{x} - s$, so $x \approx 70 - 10 = 60$

What is the standard deviation?

Estimate σ for curve A and for curve B.



A: $\sigma \approx 5$

B: $\sigma \approx 0.5$

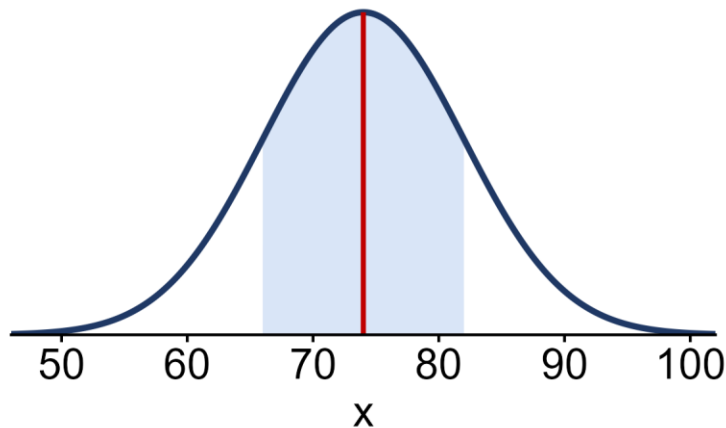
Example: using the empirical rule

A variable x has the following distribution.

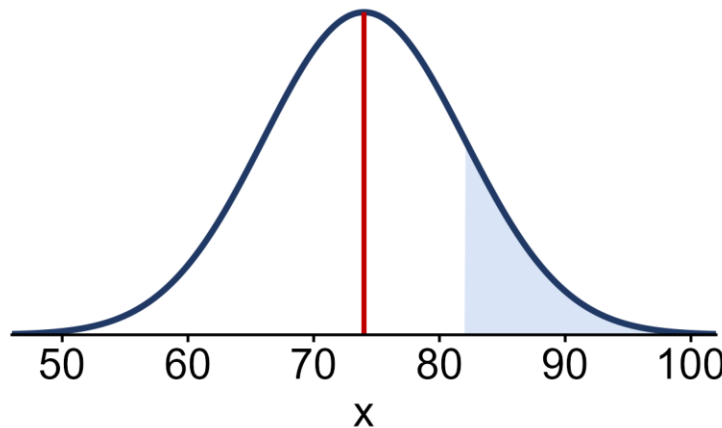
$$\bar{x} = 74 \quad s = 8$$

What percentage of the values of x falls in each shaded region?

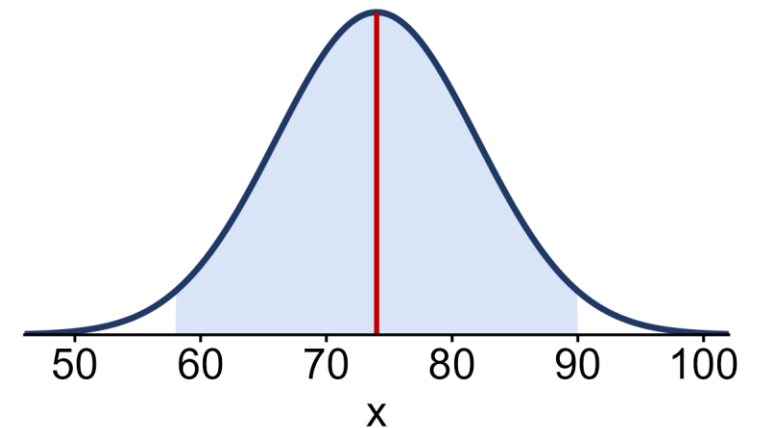
(a) 66 to 82



(b) 82 and above



(c) 58 to 90



(a) about 68%

(b) about 16%

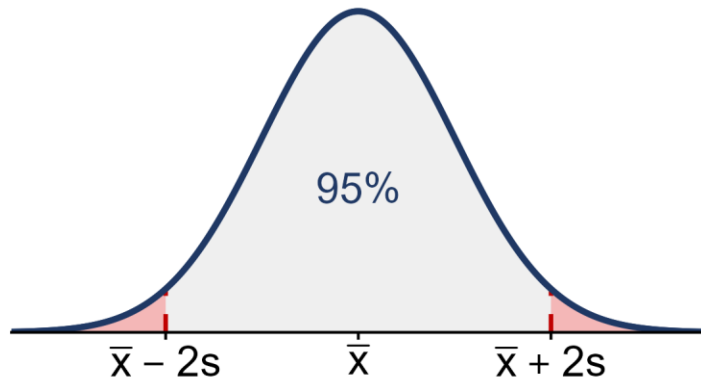
(c) about 95%

Outliers: the two-standard-deviation rule

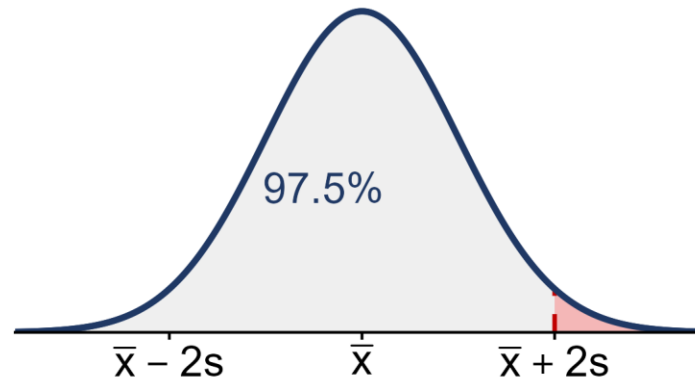
outlier

a value at least 2 standard deviations from the mean, $|z| \geq 2$ (about 5% of bell-shaped data)

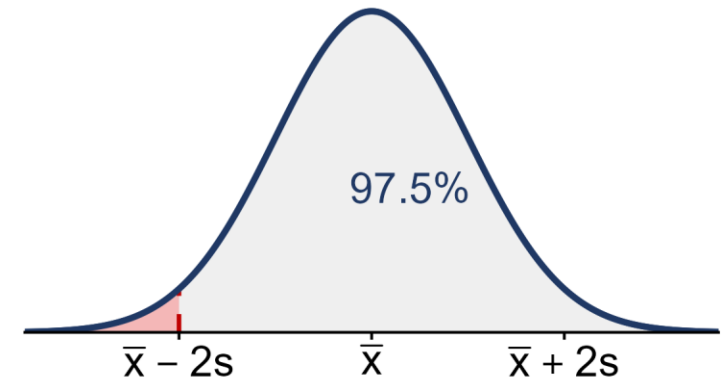
both tails: $2.5\% + 2.5\% = 5\%$



upper tail only: 2.5%



lower tail only: 2.5%



Data: student heights

A survey of 379 college students recorded each student's height and gender.

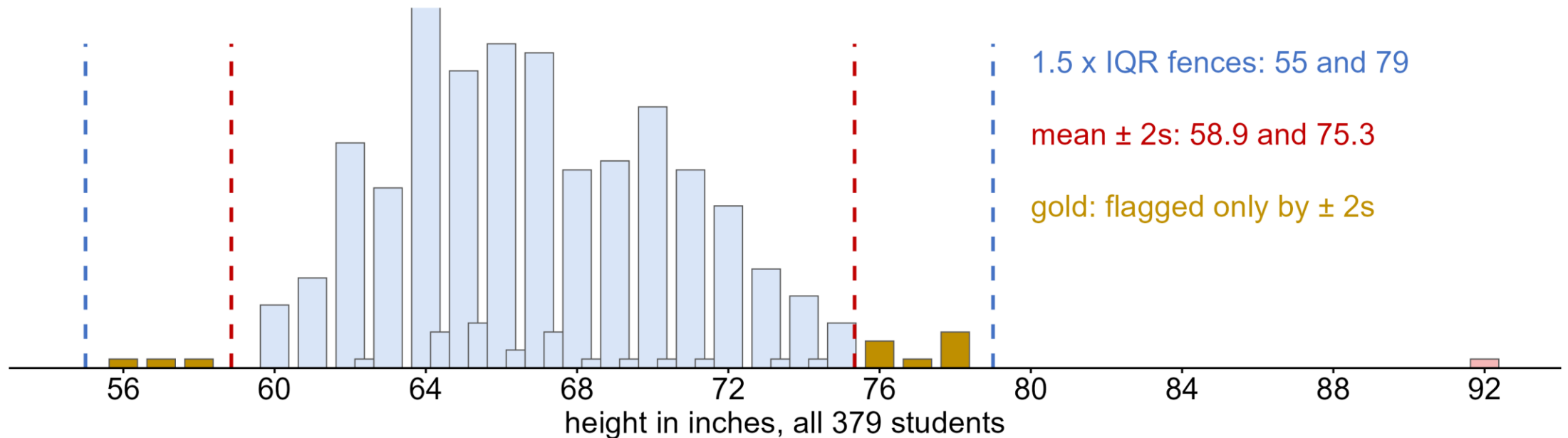
X = the height of a student, in inches

HEIGHT	GENDER
56	Female
57	Female
58	Female
60	Female
60	Female
...	...

n = 379 students (the file lists the women first, then the men, each sorted by height)

When the two outlier rules disagree

Do the $1.5 \times \text{IQR}$ rule and the $\pm 2s$ rule flag the same students?



No: the IQR rule flags only 1 student (92 in), the $\pm 2s$ rule flags 12

Why: $\pm 2s$ flags about 5% of any bell-shaped data; $1.5 \times \text{IQR}$ flags under 1%

Computing quantiles from a normal distribution

Heights of all 379 students are roughly bell shaped.

$\bar{x} = 67.1$ in $s = 4.12$ in one student is 72 in

- (a) What is the student's z-score?
- (b) What percentage of the students are shorter?

- (a) $z = 1.19$
- (b) The empirical rule cannot say: 1.19 is not a whole number of standard deviations

How it used to be done: the Z-table

the standard normal table

each entry is the area to the LEFT of z

z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545

$z = 1.19$: row 1.1, column .09 → 0.8830, so about 88% are shorter

Today: the computer does the lookup

From a value to an area: `pnorm`

`pnorm(1.19) = 0.8830`

From an area to a value: `qnorm`

`qnorm(0.90) = 1.28`

No R in front of you?

The normal calculator at stat-tools.jmks-homelab.dev does both.

A note about transformations of variables

Changing units

We often need to change the units of a variable: °F to °C, feet to meters, dollars to euros.

Linear transformations: adding, subtracting, multiplying, dividing

$y = ax + b$ (a scales, b shifts). The z-score is a linear transformation. They preserve the shape of the distribution.

Nonlinear transformations: squaring, taking roots, logarithms, exponentiation

They do NOT preserve the shape of the distribution.

What a linear transformation does to the summaries

The center moves with the whole transformation

$\bar{y} = a\bar{x} + b$, and the same for the median.

The spread ignores the shift

s and IQR are multiplied by $|a|$; b has no effect.

The shape does not change at all

Skewness, gaps and outliers all survive unchanged.

Example: inches to centimeters

The 379 heights are converted from inches to centimeters: $y = 2.54x$.

$\bar{x} = 67.1$ in $s = 4.12$ in the 72-inch student = 182.9 cm

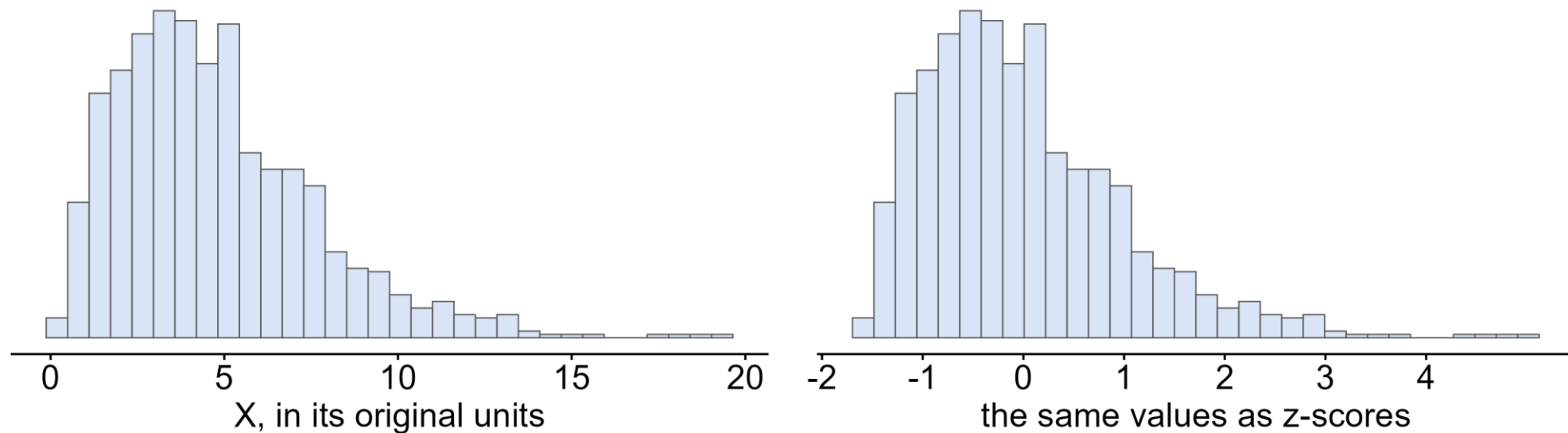
- (a) What are the mean and standard deviation in centimeters?
- (b) What is the student's z-score in centimeters?

(a) $\bar{y} = 170.4$ cm, $s = 10.46$ cm

(b) $z = 1.19$, unchanged

Does a z-score make data normal?

Does converting skewed data to z-scores make it normal?

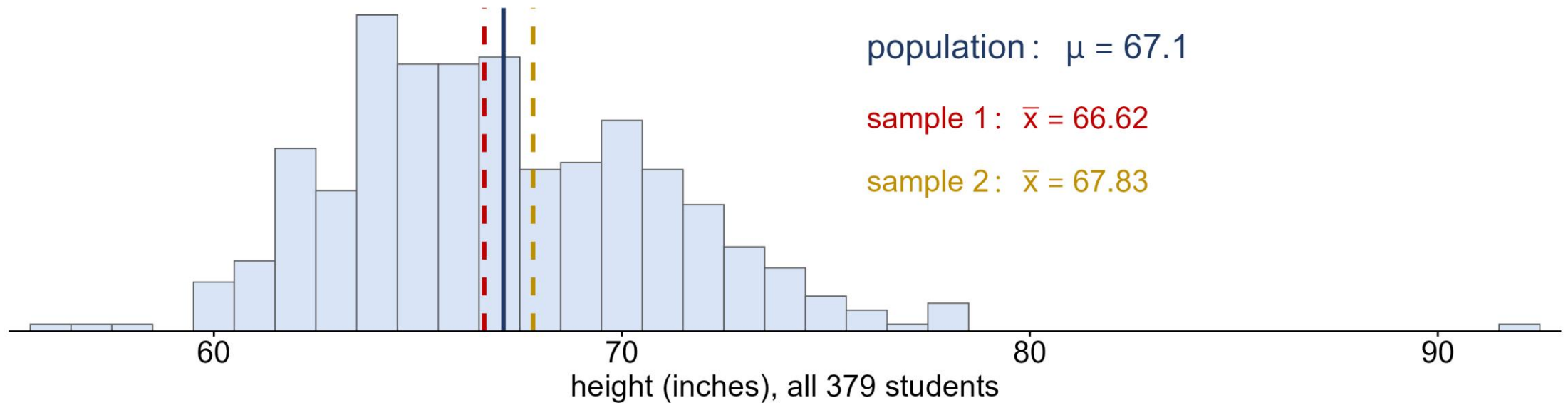


No: standardizing changes the axis, not the shape

Estimating a mean from a sample

Treat all 379 students as the population, and estimate μ from a random sample of $n = 20$.

Why do the two samples miss μ , and by different amounts?

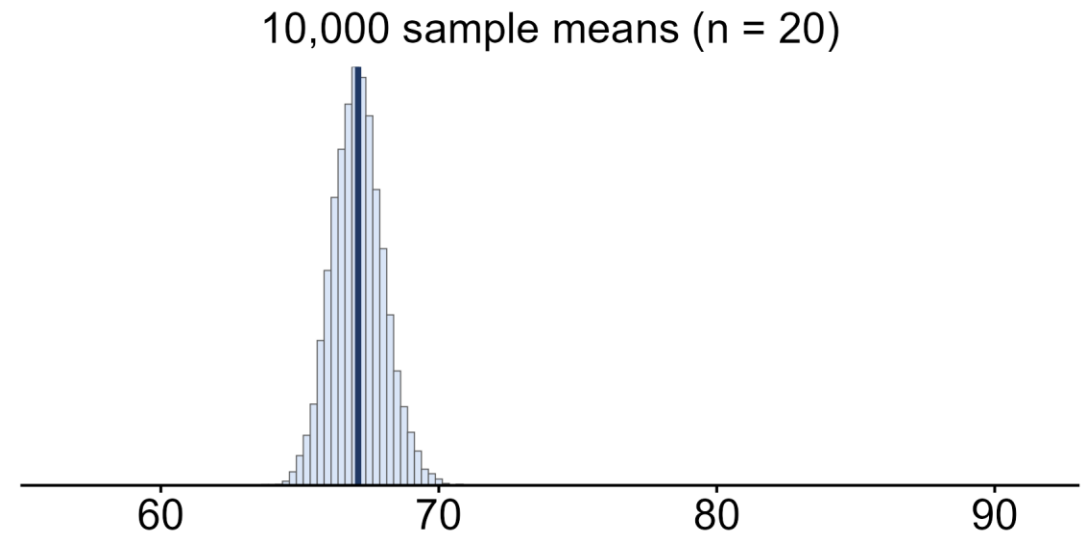
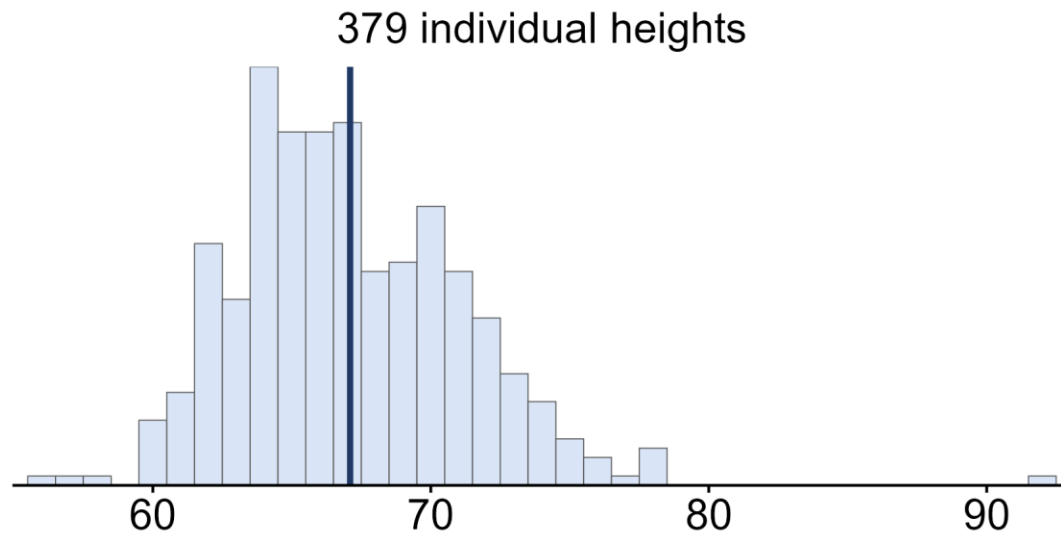


Each sample holds different students, so each $\bar{x}$ misses by a different amount: sampling error

Sampling distributions

sampling distribution

the distribution of a statistic, such as $\bar{x}$, over all possible samples of the same size n



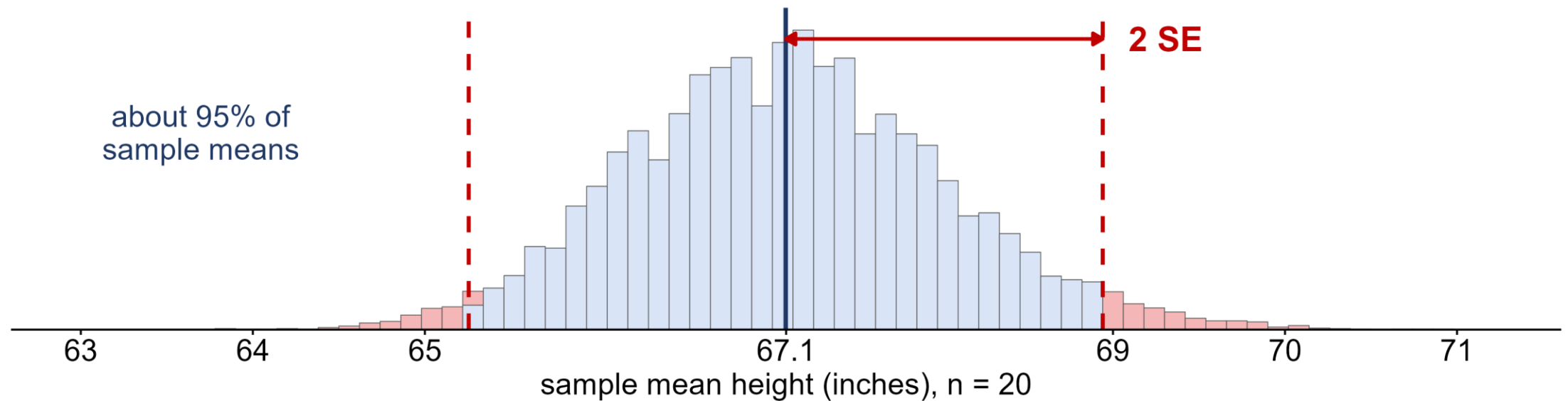
You've seen these before:

$SE = s / \sqrt{n}$ (Week 3) is its spread; the Rec Center histograms (Week 4) were sampling distributions

Margin of error

margin of error

how far we expect an estimate to fall from the parameter: the largest distance to an estimate that is not an outlier, $2 \times \text{SE}$



Example: the margin of error for $\bar{x}$

Random samples of $n = 20$ from all 379 students.

$\sigma = 4.12$ in $n = 20$

What are the standard error and the margin of error of $\bar{x}$?

$$SE = \sigma / \sqrt{n} = 4.12 / \sqrt{20} \approx 0.92 \text{ in}$$

Margin of error = $2 \times 0.92 \approx 1.8$ in: about 95% of samples land within 67.1 ± 1.8 in

Statistical significance

Someone hands you 20 students and says they are a random sample of all 379.

$\bar{x} = 70.33$ in $\mu = 67.1$ in $SE = 0.92$ in

Is this result statistically significant?

$z = (70.33 - 67.1) / 0.92 \approx 3.5$: far outside the margin of error, so yes

Statistically significant: this was not a random sample of all students (they were all men)

What you should know: the normal distribution

The two parameters

$x \sim N(\mu, \sigma)$: μ locates the center, σ sets the spread.

The empirical rule

68%, 95% and 99.7% within 1, 2 and 3 standard deviations (bell-shaped only).

The z-score

$z = (x - \bar{x}) / s$ counts standard deviations from the mean; $|z| \geq 2$ flags an outlier.

Why probability?

The margin of error said: about 95% of samples land within 2 SE of μ .

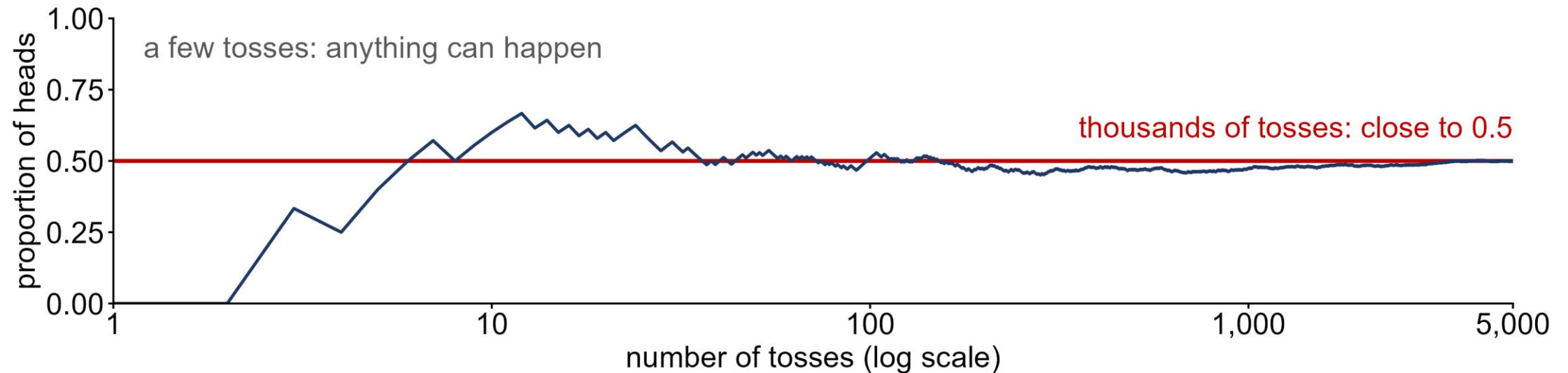
But we only ever take one sample. So what does "95%" mean?

It describes the long run: over many repeated samples, about 95% would land that close
A statement about the long run is a statement about probability

Probability

probability

the proportion of times an outcome would occur over a very long series of repetitions;
always between 0 and 1



The law of large numbers: over many repetitions, the proportion settles on the probability
Karl Pearson tossed a coin 24,000 times: 12,012 heads, a proportion of 0.5005

The language of probability

chance experiment

a process whose result cannot be predicted in advance, such as rolling a die

sample space S

the set of all possible outcomes: for one roll of a die, $S = \{1, 2, 3, 4, 5, 6\}$

event

any collection of outcomes, written with a capital letter: $A = \{\text{roll an even number}\} = \{2, 4, 6\}$

Equally likely outcomes

Roll a fair six-sided die: all six outcomes are equally likely.

$P(A) = (\text{number of outcomes in } A) / (\text{number of outcomes in } S)$

(a) What is $P(\text{even})$?

(b) What is $P(\text{not a six})$?

(a) $3 / 6 = 0.5$

(b) $5 / 6 \approx 0.83$

Example: probabilities from real data

Pick one of the 379 students from the heights data at random.

	At least 70 in	Under 70 in	Total
Female	19	243	262
Male	87	30	117
Total	106	273	379

(a) What is $P(\text{the student is at least 70 in})$?

(b) What is $P(\text{the student is female})$?

(a) $106 / 379 \approx 0.28$

(b) $262 / 379 \approx 0.69$